Borel Measurable Sets
A Summary of Definitions and Properties

Contents

1 Introduction

Borel measurable sets are among the most fundamental objects in modern probability theory, real analysis, and measure theory.

They provide the collection of subsets on which probability measures and Lebesgue measure are first defined. Every probability density function, distribution function, and measurable random variable is ultimately built upon the Borel σ-algebra.

Throughout this document, we work on the Euclidean space

$$\mathbb{R}^n.$$

2 Open and Closed Sets

Everything begins with the notion of an open set.

2.1 Open Sets

A subset

$$U \subseteq \mathbb{R}^n$$

is called **open** if for every point

$$x \in U,$$

there exists some radius

$$r > 0$$

such that

$$B_r(x) = \{y \in \mathbb{R}^n : \|x - y\| < r\} \subseteq U.$$

In words, every point has a small neighborhood completely contained in the set.

Examples include

$$(a, b), \qquad (-\infty, c), \qquad \mathbb{R}^n.$$

2.2 Closed Sets

A subset

$$F \subseteq \mathbb{R}^n$$

is called **closed** if its complement is open.

Examples include

$$[a, b], \qquad [c, \infty), \qquad \emptyset.$$

3 The Borel σ-Algebra

3.1 Definition

The collection of all Borel measurable sets is called the **Borel σ-algebra**.

It is denoted by

$$\boxed{\mathcal{B}(\mathbb{R}^n).}$$

It is defined as

$$\boxed{\mathcal{B}(\mathbb{R}^n) = \sigma(\text{all open sets}).}$$

This means that it is the smallest σ-algebra containing every open subset of $\mathbb{R}^n$.

4 What is a σ-Algebra?

A collection

$$\mathcal{F}$$

of subsets of a set

$$\Omega$$

is called a **σ-algebra** if it satisfies the following three properties.

1. Contains the Whole Space

$$\boxed{\Omega \in \mathcal{F}.}$$

Consequently,

$$\emptyset \in \mathcal{F}.$$

2. Closed Under Complements

Whenever

$$A \in \mathcal{F},$$

then

$$\boxed{A^c \in \mathcal{F}.}$$

3. Closed Under Countable Unions

If

$$A_1, A_2, \ldots \in \mathcal{F},$$

then

$$\boxed{\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.}$$

5 Immediate Consequences

From the axioms above one can prove that every σ-algebra is also closed under

Countable Intersections

Using De Morgan's Laws,

$$\boxed{\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}.}$$

Finite Unions

$$A \cup B \in \mathcal{F}.$$

Finite Intersections

$$A \cap B \in \mathcal{F}.$$

Set Differences

$$\boxed{A \setminus B = A \cap B^c \in \mathcal{F}.}$$

6 Equivalent Generators

Although the Borel σ-algebra is defined using open sets, it can be generated from many different collections.

Each of the following generates exactly the same σ-algebra:

$$\boxed{\mathcal{B}(\mathbb{R}) = \sigma(\text{open sets})}$$

$$= \sigma(\text{closed sets})$$

$$= \sigma(\text{open intervals})$$

$$= \sigma(\text{closed intervals})$$

$$= \sigma((-\infty, a))$$

$$= \sigma((-\infty, a])$$

$$= \sigma((a, \infty))$$

$$= \sigma([a, \infty)).$$

Thus, any one of these families may be used to define the Borel σ-algebra.

7 Examples of Borel Sets

Examples include

$$(a, b),$$

$$[a, b],$$

$$(a, b],$$

$$[a, b),$$

finite unions of intervals,
countable unions,
countable intersections,
singletons

$$\{x\},$$

finite sets,
countable sets,
the integers

$$\mathbb{Z},$$

the rational numbers

$$\mathbb{Q},$$

the irrational numbers

$$\mathbb{R} \setminus \mathbb{Q},$$

and every open or closed subset of

$$\mathbb{R}^n.$$

8 Examples of Non-Borel Sets

The Borel σ-algebra is extremely large; it contains every open set, every closed set, every interval, every countable set, and every set that can be obtained from these through countable unions, countable intersections, and complements.

Nevertheless,

$$\boxed{\mathcal{B}(\mathbb{R}) \neq \mathcal{P}(\mathbb{R}),}$$

where

$$\mathcal{P}(\mathbb{R})$$

denotes the power set of the real numbers.

In other words, there exist subsets of the real line that are not Borel measurable.

These sets cannot be explicitly written down by a finite formula; their existence relies on the Axiom of Choice.

We now discuss three classical examples.

8.1 Vitali Sets

Perhaps the most famous example is the *Vitali set*.

Step 1. Define an Equivalence Relation Define a relation on $\mathbb{R}$ by

$$x \sim y \iff x - y \in \mathbb{Q}.$$

This is an equivalence relation because

1. $x - x = 0 \in \mathbb{Q}$,
2. if $x - y \in \mathbb{Q}$, then

$$y - x = -(x - y) \in \mathbb{Q},$$

3. if

$$x - y \in \mathbb{Q}, \qquad y - z \in \mathbb{Q},$$

then

$$x - z = (x - y) + (y - z) \in \mathbb{Q}.$$

Thus,

$$\mathbb{R}$$

is partitioned into equivalence classes.

Step 2. Choose One Representative Using the Axiom of Choice, choose exactly one point from every equivalence class.

The resulting set is called a Vitali set,

$$V.$$

Step 3. Main Properties Every real number differs from exactly one element of

$$V$$

by a rational number.

Why Every Real Number Differs from Exactly One Element of V by a Rational Number Recall the equivalence relation

$$x \sim y \iff x - y \in \mathbb{Q}.$$

This relation partitions the real line into disjoint equivalence classes.

For every real number x, its equivalence class is

$$[x] = \{\, x + q : q \in \mathbb{Q} \,\}.$$

Thus, two real numbers belong to the same equivalence class if and only if their difference is rational.

Using the Axiom of Choice, we construct a set

$$V \subseteq \mathbb{R}$$

by selecting exactly one representative from each equivalence class.

That is,

$$\boxed{\text{For every equivalence class } [x], \text{ there exists exactly one } v \in V \text{ such that } v \in [x].}$$

This single property immediately implies the following important result.

Existence Let

$$x \in \mathbb{R}$$

be arbitrary.

Since the equivalence classes partition the real numbers,

$$x$$

belongs to exactly one equivalence class,

$$[x].$$

By construction,

$$V$$

contains one representative of this class.
Call this representative

$$v.$$

Because both

$$x \quad \text{and} \quad v$$

belong to the same equivalence class,

$$x \sim v.$$

Therefore,

$$x - v \in \mathbb{Q}.$$

Equivalently,

$$\boxed{x = v + q}$$

for some rational number

$$q \in \mathbb{Q}.$$

Hence every real number differs from at least one element of V by a rational number.

Uniqueness Suppose that there exist two elements

$$v_1, v_2 \in V$$

such that

$$x - v_1 \in \mathbb{Q}, \qquad x - v_2 \in \mathbb{Q}.$$

Subtracting the two relations gives

$$v_1 - v_2 = (v_1 - x) + (x - v_2).$$

Since the rational numbers are closed under addition,

$$v_1 - v_2 \in \mathbb{Q}.$$

Therefore,

$$v_1 \sim v_2.$$

Thus,

$$v_1 \quad \text{and} \quad v_2$$

belong to the same equivalence class.
However, the set

$$V$$

was constructed to contain exactly one representative from each equivalence class.
Consequently,

$$\boxed{v_1 = v_2.}$$

This proves that the representative is unique.

Combining existence and uniqueness, we conclude that

$$\boxed{\forall x \in \mathbb{R},\ \exists!\, v \in V \text{ such that } x - v \in \mathbb{Q}.}$$

The symbol

$$\exists!$$

means "there exists exactly one."
Equivalently, every real number can be written uniquely in the form

$$\boxed{x = v + q, \qquad v \in V,\ q \in \mathbb{Q}.}$$

where v is the unique representative of the equivalence class containing x.
Consequently,

$$\mathbb{R} = \bigcup_{q \in \mathbb{Q}} (V + q),$$

where

$$V + q = \{v + q : v \in V\}.$$

Moreover,

$$(V + q_1) \cap (V + q_2) = \varnothing, \qquad q_1 \neq q_2.$$

Step 4. Why It Is Not Borel Every Borel set is Lebesgue measurable.
However, one proves that no Vitali set is Lebesgue measurable.
Therefore,

$$\boxed{V \notin \mathcal{B}(\mathbb{R}).}$$

8.2 Bernstein Sets

A second important example is the Bernstein set.

A subset

$$B \subseteq \mathbb{R}$$

is called a Bernstein set if

1. every nonempty perfect set intersects B,
2. every nonempty perfect set also intersects

$$\mathbb{R} \setminus B.$$

Thus,

$$B$$

contains no entire perfect set, but neither does its complement.

Existence Bernstein sets exist only because of the Axiom of Choice.
Their construction requires a transfinite induction over all perfect subsets of the real line.

Why They Are Not Borel A fundamental theorem of descriptive set theory states:

Every uncountable Borel subset of $\mathbb{R}$ *contains a nonempty perfect subset.*

If a Bernstein set were Borel, then it would contain a perfect subset.
This contradicts its defining property.
Hence,

$$\boxed{B \notin \mathcal{B}(\mathbb{R}).}$$

8.3 Hamel Bases

A Hamel basis is a basis of the vector space

$$\mathbb{R}$$

over the field

$$\mathbb{Q}.$$

Thus,
every real number admits a unique finite representation

$$x = q_1 b_1 + \cdots + q_n b_n,$$

where

$$q_i \in \mathbb{Q}$$

and

$$b_i$$

belong to the Hamel basis.

Existence Every vector space possesses a basis.

The proof uses Zorn's Lemma, which is equivalent to the Axiom of Choice.

Why Hamel Bases Are Not Borel A deep theorem states that every Hamel basis is non-Lebesgue measurable.

Since every Borel set is Lebesgue measurable, it follows immediately that

$$\boxed{\text{Every Hamel basis is not Borel measurable.}}$$

The proof relies on the Steinhaus Theorem and the algebraic structure of Hamel bases.

8.4 Summary

These examples illustrate an important fact.

Although the Borel σ-algebra contains virtually every set encountered in analysis, it is still strictly smaller than the collection of all subsets of the real numbers.

The existence of non-Borel sets depends essentially on the Axiom of Choice and cannot be demonstrated constructively.

The three classical examples are summarized below.

Set	**Exists by**	**Reason it is not Borel**
Vitali set	Axiom of Choice	Not Lebesgue measurable
Bernstein set	Transfinite induction	Contains no perfect subset
Hamel basis	Zorn's Lemma	Not Lebesgue measurable

9 Why Are Borel Sets Important?

Probability measures are defined on Borel sets.

If

$$X$$

is a random variable, then for every Borel set

$$B,$$

the probability

$$P(X \in B)$$

is well defined.

Likewise,

Lebesgue measure,

Gaussian measures,

Poisson processes,
Brownian motion,
and nearly every object in probability theory are defined on Borel σ-algebras.

10 Borel Measurable Functions

A function

$$f : \mathbb{R}^n \to \mathbb{R}$$

is called **Borel measurable** if

$$\boxed{f^{-1}(B) \in \mathcal{B}(\mathbb{R}^n)}$$

for every Borel set

$$B \subseteq \mathbb{R}.$$

Equivalently,

$$f^{-1}((-\infty, a))$$

must be a Borel set for every real number

$$a.$$

Every continuous function is Borel measurable.

11 Summary of Properties

Property	Holds?
Contains every open set	Yes
Contains every closed set	Yes
Closed under complements	Yes
Closed under countable unions	Yes
Closed under countable intersections	Yes
Closed under finite unions	Yes
Closed under finite intersections	Yes
Contains every interval	Yes
Contains every singleton	Yes
Contains every countable set	Yes
Contains every continuous preimage	Yes
Contains every subset of $\mathbb{R}$	No

12 Key Facts to Remember

1. The Borel σ-algebra is generated by the open sets.
2. It is the smallest σ-algebra containing all open sets.
3. Every open, closed, and interval-type set is Borel measurable.
4. Every continuous function is Borel measurable.
5. Every probability measure on $\mathbb{R}^n$ is defined on Borel sets.
6. Random variables are measurable precisely so that probabilities of the form

$$P(X \in B)$$

are well defined for every Borel set

$$B.$$